Foatic Homomesy

Telescoping proofs for orbit averages on S_n

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Homomesy

A teaser: orbits in S_7 under a strange action

Let $\tau = \mathcal{I} \circ \mathcal{F}^{-1} \circ \mathcal{C} \circ \mathcal{F}$ act on S_7 (each piece defined shortly). Two of its 288 orbits, written in cycle notation:

orbit of size 4	fix	orbit of size 6	fix
$(5,4)(6,3)(7,1,2)$	0	$(4)(5,3)(6,1,2)(7)$	2
$(3)(4,2)(5,1)(7,6)$	1	$(4,3)(5,2)(7,1,6)$	0
$(5,4)(6,3)(7,2,1)$	0	$(4)(5,3)(6,1)(7,2)$	1
$(3)(4,2)(5,1)(6)(7)$	3	$(4,3)(5,2)(7,6,1)$	0
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average	1	average	1

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All 288 orbits (sizes from 2 to 80) average to 1, for every n: homomesy.

Homomesy (Propp–Roby, 2015)

Definition

Let τ be an invertible map on a finite set X and $s : X \to \mathbb{C}$ a statistic. Then s is c-mesic under τ if

$$\frac{1}{|\mathcal{O}|} \sum_{x \in \mathcal{O}} s(x) \;=\; c \qquad \text{for every } \tau\text{-orbit } \mathcal{O} \subseteq X.$$

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The average over an orbit equals the global average, no matter which orbit.

In the teaser: fix is 1-mesic under $\mathcal{I} \circ \mathcal{F}^{-1} \circ \mathcal{C} \circ \mathcal{F}$ on S_n, for every n.

Permutation homomesies so far: Elder–Lafrenière–McNicholas–Striker–Welch (FindStat survey, 2024), Dowling–Lafrenière (toggling actions, 2025), and the Foatic maps of La Croix–Roby and Sheridan-Rossi (this talk).

Foatic maps

CCD and the Foata bijection $\mathcal{F}$

Canonical cycle decomposition (CCD): each cycle starts with its largest element; cycles are listed by increasing maxima. E.g. $\sigma = 417682953$ has CCD $(6,2,1,4)(8,5)(9,3,7)$.

Definition (Rényi–Foata; Stanley's "fundamental bijection")

$\mathcal{F} : S_n \to S_n$ writes a permutation in CCD, drops the parentheses, and reads the resulting word as a one-line notation (OLN).

CCD of σ: `(6,2,1,4)(8,5)(9,3,7)`

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OLN of $\mathcal{F}(\sigma)$: `6 2 1 4 8 5 9 3 7`

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The inverse $\mathcal{F}^{-1}$: insert a left parenthesis before each left-to-right maximum of the OLN; these become the cycle heads.

$$\mathcal{F}^{-1}(621485937) = (6,2,1,4)(8,5)(9,3,7) \qquad \text{(LtR maxima } 6,8,9).$$

Not to be confused with: the other Foata bijection

First fundamental transformation $\mathcal{F}$
(Rényi 1962; Foata–Schützenberger 1970)

- CCD $\to$ OLN by dropping parentheses.
- Cycle heads become LtR maxima; $\mathrm{anxc}(\pi) = \mathrm{des}(\mathcal{F}\pi)$, so exc $\sim$ des.
- This talk. Elder et al. ("first fundamental transform"): no homomesic statistics.

Second fundamental transformation Φ
(Foata 1968; Foata–Schützenberger 1978)

- OLN $\to$ OLN, built letter by letter by cyclically shifting blocks.
- $\mathrm{maj}(w) = \mathrm{inv}(\Phi w)$.
- FindStat's "Foata bijection" (Map 67). Elder et al.: $\mathrm{maj} - \mathrm{inv}$ is homomesic.

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Example ($w = 31542$, CCD $(4)(5,2,1,3)$):

$$\mathcal{F}(w) = 45213, \qquad \Phi(w) = 53412 \quad (\mathrm{maj}(w) = 8 = \mathrm{inv}(\Phi w)).$$

The five dihedral involutions

Think of σ as its permutation matrix (points $(i, \sigma(i))$ in an $n \times n$ grid). The symmetry group of the square acts; its five elements of order 2 are:

$\mathcal{C}$	complement	$\sigma(i) \mapsto n + 1 - \sigma(i)$	(flip top–bottom)
$\mathcal{R}$	reversal	reverse the OLN	(flip left–right)
$\mathcal{I}$	inverse	$\sigma \mapsto \sigma^{-1}$	(main diagonal)
$\mathcal{Q}^2 = \mathcal{C}\mathcal{R} = \mathcal{R}\mathcal{C}$	rotation	reverse-complement	(180° rotation)
$\mathcal{D} = \mathcal{Q}^2\mathcal{I}$	rotated inverse		(anti-diagonal)

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On 621485937 **($n = 9$):** $\mathcal{C} \mapsto 489625173$, $\mathcal{R} \mapsto 739584126$, $\mathcal{Q}^2 \mapsto 371526984$.

Foatic maps

Definition (La Croix–Roby; Sheridan-Rossi)

A Foatic map is a composition

$$S_n \xrightarrow{\mathcal{F}} S_n \xrightarrow{\mathcal{A}} S_n \xrightarrow{\mathcal{F}^{-1}} S_n \xrightarrow{\mathcal{B}} S_n$$

with $\mathcal{A}, \mathcal{B}$ dihedral involutions. We call it "$\mathcal{A}$-$\mathcal{B}$".

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Example. The teaser map $\mathcal{I} \circ \mathcal{F}^{-1} \circ \mathcal{C} \circ \mathcal{F}$ is complement-inversion, $\mathcal{C}$-$\mathcal{I}$.

There are $5 \times 5 = 25$ Foatic maps. La Croix–Roby and Sheridan-Rossi found (and proved, or conjectured) many homomesies for them.

Contrast: Elder et al. (2024) searched FindStat and found no statistic homomesic under $\mathcal{F}$ itself.

Computing the teaser by hand

$\tau = \mathcal{I} \circ \mathcal{F}^{-1} \circ \mathcal{C} \circ \mathcal{F}$, writing each permutation in whichever notation the next step reads:

π (CCD)	$\xrightarrow{\mathcal{F}}$ OLN	$\xrightarrow{\mathcal{C}}$ OLN	$\xrightarrow{\mathcal{F}^{-1}}$ CCD	$\xrightarrow{\mathcal{I}}$ CCD
$(5,4)(6,3)(7,1,2)$	5463712	3425176	$(3)(4,2)(5,1)(7,6)$	$(3)(4,2)(5,1)(7,6)$

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Results

The four main theorems

	Statistic	Foatic action(s)	c
Thm 3.1	fix	$\mathcal{C}$-$\mathcal{I}$, $\mathcal{C}$-$\mathcal{Q}^2$, $\mathcal{R}$-$\mathcal{Q}^2$, $\mathcal{R}$-$\mathcal{I}$	1
Thm 3.2	$\overleftarrow{\max} - \overleftarrow{\min}$	$\mathcal{Q}^2$-$\mathcal{C}$	0
Thm 3.3	$D_1 - D_{n-1}$ $(n \geq 2)$	$\mathcal{Q}^2$-$\mathcal{R}$	0
Thm 3.4	$\mathrm{fix}_1 - \mathrm{fix}_n$	$\mathcal{D}$-$\mathcal{R}$, $\mathcal{R}$-$\mathcal{D}$, $\mathcal{C}$-$\mathcal{D}$, $\mathcal{D}$-$\mathcal{C}$	0

- $\overleftarrow{\max}, \overleftarrow{\min}$: number of right-to-left maxima / minima of the OLN.
- $D_k(\sigma) = [\sigma(k) > \sigma(k+1)]$ (Iverson bracket: 1 if true, 0 if false).
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All four are proved by one method: find an explicit telescoping function.

Where these come from

Status before this work.

- fix under $\mathcal{R}$-$\mathcal{I}$: proved by La Croix–Roby (2020, Thm. 14).
- fix under $\mathcal{C}$-$\mathcal{I}$, $\mathcal{C}$-$\mathcal{Q}^2$, $\mathcal{R}$-$\mathcal{Q}^2$: La Croix–Roby Conjectures 16, 17.1, 18.
- Theorems 3.2–3.4: Sheridan-Rossi (PhD thesis, UConn 2020), Conjectures 2.5.4–2.5.6.

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What's new.

- One uniform proof strategy, building on Propp's coboundary criterion: an explicit telescoping function for every one of these homomesies.
- *Compatibility*: one telescoping function can serve several actions at once, so mixing actions still gives constant averages.
- Bijective identities across $\mathcal{F}$, including an unexpected invariance of $\underleftarrow{\min}$ under $\mathcal{F}^{-1} \circ \mathcal{Q}^2 \circ \mathcal{F}$.
- Small n is less trustworthy here. Elder et al.'s FindStat sweep found no false positives using data up to $n = 6$. We find two Foatic "homomesies" that hold for all $n \leq 8$ and first fail at $n = 9$.

Telescoping

The telescoping lemma

Lemma 4.1

Let τ be an action on X, $s : X \to \mathbb{C}$ and $c \in \mathbb{C}$. If some $f : X \to \mathbb{C}$ satisfies

$$s(x) \;=\; f(x) - f(\tau(x)) + c \qquad \text{for all } x \in X,$$

then s is c-mesic under τ.

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This generalizes Propp's coboundary criterion for 0-mesy (*A spectral theory for combinatorial dynamics*, Prop. 2.4); Defant–Hopkins–Poznanović–Propp (2023) use a similar device.

Telescoping on the teaser

For $\mathcal{C}$-$\mathcal{I}$, take $f = -\operatorname{exc}$. Along the orbit of size 4, check that $s(x) = f(x) - f(\tau x) + 1$:

x	$s(x) = \operatorname{fix}(x)$	$f(x) = -\operatorname{exc}(x)$	$f(x) - f(\tau x) + 1$
$(5,4)(6,3)(7,1,2)$	0	-4	$-4+3+1=0$
$(3)(4,2)(5,1)(7,6)$	1	-3	$-3+3+1=1$
$(5,4)(6,3)(7,2,1)$	0	-3	$-3+2+1=0$
$(3)(4,2)(5,1)(6)(7)$	3	-2	$-2+4+1=3$

Summing around the orbit, the f-terms cancel, leaving 1 per step.

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Summing around the orbit, the f-terms cancel, leaving 1 per step.

The point: we never have to understand the orbits. We only need a *local*, one-step identity between π and $\tau(\pi)$.

The converse holds, so the work is finding f

Remark 4.2

If s is c-mesic under τ, then a telescoping f always exists, and it is unique up to adding a constant on each orbit.

Why: on an orbit $x_1 \mapsto \cdots \mapsto x_k \mapsto x_1$, choose $f(x_1)$ freely and solve $f(x_{i+1}) = f(x_i) - s(x_i) + c$. The last equation closes up exactly because the s-values on the orbit sum to kc.

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Equivalently: s is c-mesic $\iff$ $s - c$ sums to 0 over every orbit $\iff$ $s - c = f - f \circ \tau$ for some f.

So existence is free. The theorem is a *closed form* for f in terms of familiar statistics.

The linear-algebra picture (Propp 2021)

τ acts on functions $X \to \mathbb{C}$ by the time-evolution operator $Ug := g \circ \tau$. As $\tau^N = \mathrm{id}$, U is diagonalizable with eigenvalues N-th roots of unity:

$$\mathbb{C}^X = \underbrace{V_1}_{=\ker(1-U)} \oplus \underbrace{\bigoplus_{\zeta \neq 1} V_\zeta}_{=\mathrm{Im}(1-U)}$$

- V_1: the *invariants*, i.e. functions constant on orbits.
- Other eigenspaces: exactly the 0-*mesic* functions, which are exactly the *coboundaries* $g - Ug$ (Props. 2.1, 2.4).
- So s is c-mesic $\iff$ $s - c \in \mathrm{Im}(1 - U)$, with f unique mod V_1: Lemma 4.1 + Remark 4.2.

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Propp's explicit preimage. On an orbit of length k, if h is 0-mesic then $g = -\frac{1}{k}\sum_{j=0}^{k-1}(j+1)\,U^j h$ satisfies $h = g - Ug$. On the teaser orbit with $h = \mathrm{fix} - 1 = (-1, 0, -1, 2)$, this gives $g = (-1, 0, 0, 1) = -\mathrm{exc} + 3$.

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Propp's g needs the whole orbit. Our telescopes are standard statistics, read off π alone.

Bonus: compatible actions

Definition and Corollary 4.4

Actions $\tau_1, \ldots, \tau_m$ are *s*-compatible via f if $s = f - f \circ \tau_i + c_i$ for every i. Then on the multigraph with edges $x \to \tau_i(x)$, every closed walk using a_i edges of type τ_i has s-average

$$\frac{\sum_i a_i c_i}{\sum_i a_i}.$$

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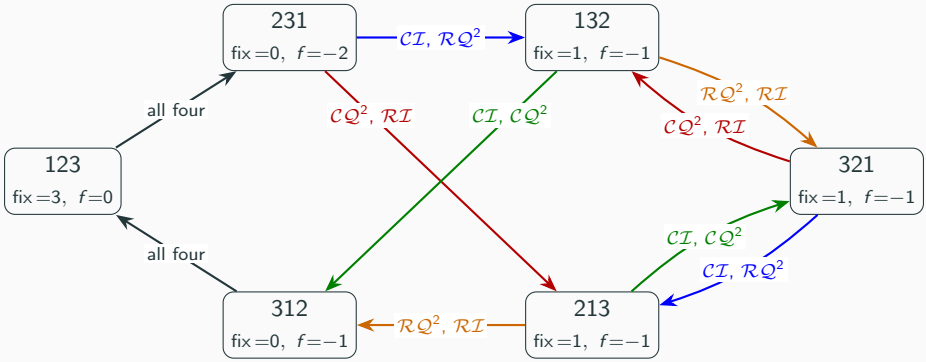
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Example. $\mathcal{C}$-$\mathcal{I}$, $\mathcal{C}$-$\mathcal{Q}^2$, $\mathcal{R}$-$\mathcal{Q}^2$, $\mathcal{R}$-$\mathcal{I}$ are all fix-compatible via $f = -\text{exc}$. So *any* closed walk that freely mixes these four maps has fix-average 1.

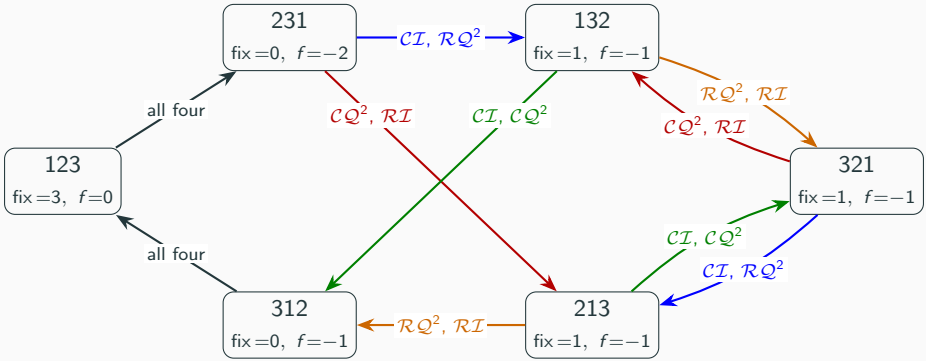
(Compare Elder et al.: linear combinations of homomesic statistics are homomesic. Compatibility is the analogous statement for *mixing actions*.)

The four fix-compatible actions on S_3



$f = -\,\mathrm{exc}$. Every edge $x \to y$ satisfies $\mathrm{fix}(x) = f(x) - f(y) + 1$, so *every* closed walk has fix-average 1.

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E.g. $123 \to 231 \to 132 \to 321 \to 213 \to 312 \to 123$: $\mathrm{fix} = 3, 0, 1, 1, 1, 0$, average 1.

The toolkit

Every proof is a chain of statistic identities

	Identity (for all $\pi, \theta \in S_n$)	Idea
Dihedral symmetries flip inequalities		
5.1	$\mathrm{exc}(\pi) = \mathrm{anxc}(\mathcal{I}\pi) = \mathrm{anxc}(\mathcal{Q}^2\pi)$	
5.3	$\mathrm{des}(\pi) = \mathrm{asc}(\mathcal{C}\pi) = \mathrm{asc}(\mathcal{R}\pi)$	
5.6	$\underleftarrow{\max}(\pi) = \underleftarrow{\min}(\mathcal{C}\pi)$	
5.20	$D_1(\mathcal{R}\theta) = 1 - D_{n-1}(\theta),\ \ D_1(\mathcal{I}\theta) + D_1(\mathcal{I}\mathcal{R}\theta) = 1$	
Crossing the Foata bijection		
5.2	$\mathrm{anxc}(\pi) = \mathrm{des}(\mathcal{F}\pi)$	Foata's exc $\sim$ des argument
5.4	$\mathrm{asc}(\mathcal{F}\pi) = \mathrm{fix}(\pi) + \mathrm{exc}(\pi) - 1$	$n = \mathrm{anxc} + \mathrm{fix} + \mathrm{exc}$
5.11	split fixed points of θ = those of $\mathcal{F}^{-1}\theta$	$\mathcal{F}$ commutes with $\oplus$
5.23–26	$\pi(n) = n \iff \mathcal{F}\pi(n) = n,\ \ \pi(1) = n \iff \mathcal{F}\pi(n) = 1, \ldots$	track one entry
Two identities for the involution $U := \mathcal{F}^{-1} \circ \mathcal{Q}^2 \circ \mathcal{F}$		
★ 5.8	$\underleftarrow{\min}(U\pi) = \underleftarrow{\min}(\pi)$	explicit bijection (next section)
★ 5.22	$D_1(\mathcal{I}\, U\theta) = D_{n-1}(\theta)$	case analysis

Everything above the last block is short. All the difficulty sits in U, and U is exactly what appears in $\mathcal{Q}^2$-$\mathcal{C} = \mathcal{C} \circ U$ and $\mathcal{Q}^2$-$\mathcal{R} = \mathcal{R} \circ U$.

Theorem 3.1 in one line

For $\tau = \mathcal{I} \circ \mathcal{F}^{-1} \circ \mathcal{C} \circ \mathcal{F}$, chain the lemmas:

$$\mathrm{exc}(\tau\pi) \overset{5.1}{=} \mathrm{anxc}(\mathcal{F}^{-1}\mathcal{C}\mathcal{F}\pi) \overset{5.2}{=} \mathrm{des}(\mathcal{C}\mathcal{F}\pi) \overset{5.3}{=} \mathrm{asc}(\mathcal{F}\pi) \overset{5.4}{=} \mathrm{fix}(\pi) + \mathrm{exc}(\pi) - 1.$$

Rearranged: $\mathrm{fix}(\pi) = (-\mathrm{exc})(\pi) - (-\mathrm{exc})(\tau\pi) + 1$. So fix is 1-mesic, with $f = -\mathrm{exc}$.

Theorem 3.1 in one line

For $\tau = \mathcal{I} \circ \mathcal{F}^{-1} \circ \mathcal{C} \circ \mathcal{F}$, chain the lemmas:

$$\mathrm{exc}(\tau\pi) \overset{5.1}{=} \mathrm{anxc}(\mathcal{F}^{-1}\mathcal{C}\mathcal{F}\pi) \overset{5.2}{=} \mathrm{des}(\mathcal{C}\mathcal{F}\pi) \overset{5.3}{=} \mathrm{asc}(\mathcal{F}\pi) \overset{5.4}{=} \mathrm{fix}(\pi) + \mathrm{exc}(\pi) - 1.$$

Rearranged: $\mathrm{fix}(\pi) = (-\mathrm{exc})(\pi) - (-\mathrm{exc})(\tau\pi) + 1$. So fix is 1-mesic, with $f = -\mathrm{exc}$.

The other three actions come for free. Lemmas 5.1 and 5.3 each have two halves:

$$\mathrm{exc} \xrightarrow{\mathcal{I} \text{ or } \mathcal{Q}^2} \mathrm{anxc} \xrightarrow{\mathcal{F}} \mathrm{des} \xrightarrow{\mathcal{C} \text{ or } \mathcal{R}} \mathrm{asc} \xrightarrow{\mathcal{F}^{-1}} \mathrm{fix} + \mathrm{exc} - 1.$$

The 2×2 choices give exactly $\mathcal{C}$-$\mathcal{I}$, $\mathcal{C}$-$\mathcal{Q}^2$, $\mathcal{R}$-$\mathcal{I}$, $\mathcal{R}$-$\mathcal{Q}^2$, all with the same $f = -\mathrm{exc}$. This is where compatibility comes from.

Theorem 3.4: tracking a single entry

$\mathsf{fix}_1 - \mathsf{fix}_n$ is 0-mesic under four actions, via two telescoping functions:

$$\begin{aligned} \mathcal{C}\text{-}\mathcal{D},\ \mathcal{D}\text{-}\mathcal{C}\text{:} \quad & f(\pi) = \mathsf{fix}_1(\pi) + [\pi(1) = n] \\ \mathcal{R}\text{-}\mathcal{D},\ \mathcal{D}\text{-}\mathcal{R}\text{:} \quad & f(\pi) = -[\pi(1) = n] - \mathsf{fix}_n(\pi) \end{aligned}$$

For $\mathcal{C}$-$\mathcal{D}$ the telescoping identity splits into $\mathsf{fix}_n(\pi) = [\tau\pi(1) = n]$ and $[\pi(1) = n] = \mathsf{fix}_1(\tau\pi)$. For example:

$$\begin{aligned} \pi(n) = n &\iff \mathcal{F}\pi(n) = n \iff \mathcal{C}\mathcal{F}\pi(n) = 1 \iff \mathcal{F}^{-1}\mathcal{C}\mathcal{F}\pi(1) = n \\ &\iff \mathcal{I}\mathcal{F}^{-1}\mathcal{C}\mathcal{F}\pi(n) = 1 \iff \mathcal{Q}^2\mathcal{I}\mathcal{F}^{-1}\mathcal{C}\mathcal{F}\pi(1) = n \iff \tau\pi(1) = n. \end{aligned}$$

(Key step: the last letter of an OLN lies in the last cycle, whose head is n.)

Each of Lemmas 5.23–5.26 is two such chains.

Theorem 3.3: everything reduces to one identity for U

$f(\pi) = D_1(\pi) - D_1(\mathcal{I}\pi)$. Since $U = \mathcal{F}^{-1}\mathcal{Q}^2\mathcal{F}$ is an involution, Lemmas 5.20–5.21 reduce the telescoping identity for $\mathcal{Q}^2\text{-}\mathcal{R} = \mathcal{R} \circ U$ to

Lemma 5.22

For $n \geq 2$ and all $\theta \in S_n$: $\quad D_1(\mathcal{I}\, U\theta) = D_{n-1}(\theta)$.

In words: 1 comes after 2 in $U\theta \iff \theta$ ends with a descent.

Proof of Lemma 5.22

Goal: $D_1(\mathcal{I}\,U\theta) = D_{n-1}(\theta)$, i.e. 1 comes after 2 in $U\theta \iff \theta(n-1) > \theta(n)$.

Generic case ($\theta(n) \neq n$ and $n-1$ is not last in its cycle): in $\mathcal{Q}^2\mathcal{F}\theta$ the letters $1, 2$ are immediately preceded by $n{+}1{-}\theta(n)$ and $n{+}1{-}\theta(n{-}1)$, so neither is a cycle head and

$$\text{position of } 1 \text{ in } U\theta = n{+}1{-}\theta(n), \qquad \text{position of } 2 = n{+}1{-}\theta(n{-}1).$$

Comparing them compares $\theta(n-1)$ with $\theta(n)$. The boundary cases are where n or $n{-}1$ ends a cycle of θ.

Proof of Lemma 5.22 (continued): boundary cases

Goal: $D_1(\mathcal{I}\, U\theta) = D_{n-1}(\theta)$, where $U = \mathcal{F}^{-1}\mathcal{Q}^2\mathcal{F}$.

(ii) $\theta(n) = n$**:** then $D_{n-1}(\theta) = 0$, and $\theta(n){=}n \iff \mathcal{F}\theta(n){=}n \iff \mathcal{Q}^2\mathcal{F}\theta(1){=}1 \iff U\theta(1){=}1$. So 1 is in position 1 of $U\theta$ and $D_1(\mathcal{I}U\theta) = 0$.

(iii) $\theta(n) \neq n$**, and** $n{-}1$ **is last in its cycle:** then $\theta(n{-}1) \in \{n{-}1, n\}$, which is $> \theta(n)$, so $D_{n-1}(\theta) = 1$.

- $\theta(n{-}1) = n{-}1$: $\mathcal{Q}^2\mathcal{F}\theta = \alpha \cdot 1 \cdot 2 \cdot \beta$ with $\alpha \neq \emptyset$, so $U\theta(1) = 2$, i.e. 1 sits in position 2.
- $\theta(n{-}1) = n$: $\mathcal{Q}^2\mathcal{F}\theta = 2 \cdot \alpha \cdot 1 \cdot \beta$. If $\alpha = \emptyset$, then $(2, 1)$ is a cycle of $U\theta$. Otherwise $U\theta(2) = 2$ and $U\theta(1) \neq 1$.

In every subcase, 1 appears after 2 in $U\theta$.

An invariance of RtL minima

An unexpected invariance

Proposition 5.8

For every $\theta \in S_n$: $\underleftarrow{\min}\big(\mathcal{F}^{-1}(\mathcal{Q}^2\theta)\big) = \underleftarrow{\min}\big(\mathcal{F}^{-1}(\theta)\big)$.

Equivalently, $\underleftarrow{\min}(U\pi) = \underleftarrow{\min}(\pi)$. With Lemma 5.6 this gives Theorem 3.2 with $f = \underleftarrow{\max}$.

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Equivalently, $\underleftarrow{\min}(U\pi) = \underleftarrow{\min}(\pi)$. With Lemma 5.6 this gives Theorem 3.2 with $f = \underleftarrow{\max}$.

Why it's surprising.

- $\mathcal{Q}^2$ alone does *not* preserve $\underleftarrow{\min}$: it turns RtL minima into LtR maxima.
- The analogous statement fails for $\mathcal{C}$, $\mathcal{R}$, $\mathcal{I}$ and $\mathcal{D}$ in place of $\mathcal{Q}^2$ (computer check, $n \leq 7$).
- The RtL minima of the two permutations sit at different positions with different values. The bijection between them is not obvious.

Example: $\theta = 1432$: $\mathcal{F}^{-1}\theta = 1423$ and $\mathcal{F}^{-1}\mathcal{Q}^2\theta = \mathcal{F}^{-1}(3214) = 3124$. Both have 3 RtL minima.

Where RtL minima of $\mathcal{F}^{-1}(\theta)$ come from

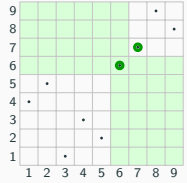
Let $\sigma = \mathcal{F}^{-1}(\theta)$ and plot σ as the points $(i, \sigma(i))$. RtL minima are the points with nothing to their southeast.

Lemma 5.17. $(i, \sigma(i))$ is a RtL minimum of σ iff it is either

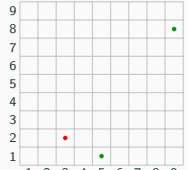
1. a split fixed point (i, i) of θ, i.e. $\theta = \alpha \oplus 1 \oplus \beta$ with $|\alpha| = i - 1$; or
2. a descent pair $(a, b) = (\theta(j), \theta(j+1))$, $a > b$, with no other descent pair (a', b') to its southeast $(b' < b < a < a')$. Call these $E^{\min}(\theta)$.

(The descent pairs of θ are exactly the antiexcedance points $(a, \sigma(a))$, $\sigma(a) < a$, of σ.)

$\theta = 451326798$



split fixed points 6, 7 (shaded: NW and SE of 6 are empty)



descent pairs $(5, 1), (9, 8) \in E^{\min}$;
$(3, 2) \notin E^{\min}$, since $(5, 1)$ is SE of it

Proof of the classification (Lemmas 5.7–5.17)

Let $\sigma = \mathcal{F}^{-1}\theta$, and let b be a RtL minimum value of σ, at position $a = \sigma^{-1}(b)$.

Pigeonhole (5.7). The $n - a$ entries to the right of b all exceed b, so $n - a \leq n - b$, i.e. $a \geq b$.

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Case $a = b$. Then b is a fixed point and a RtL minimum of σ, hence also a LtR maximum: any two of {fixed point, LtR max, RtL min} imply the third (5.10). So $\sigma = \alpha \oplus 1 \oplus \beta$, and since $\mathcal{F}$ commutes with $\oplus$, b is a split fixed point of θ (5.11).

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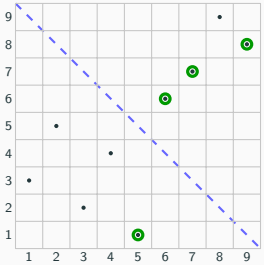
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Case $a > b$. Then (a, b) is an antiexcedance point of σ, i.e. a descent pair $a\,b$ of θ (5.14). It is a RtL minimum iff no point of σ lies to its SE. But any point (a', b') SE of it has $a' > a > b > b'$, so it is *also* an antiexcedance point, i.e. a descent pair. Hence "no point SE" $\iff$ "no descent pair SE" $\iff$ $(a, b) \in E^{\min}(\theta)$ (5.16). $\square$

The bijection: reflect across the anti-diagonal

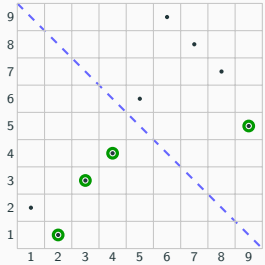


$\sigma = \mathcal{F}^{-1}(\theta) = 352416798$

RtL minima $(5,1), (6,6), (7,7), (9,8)$

$\rho(a,b)=(n+1-b,\ n+1-a)$

$$(5,1) \mapsto (9,5)$$
$$(9,8) \mapsto (2,1)$$
$$(6,6) \mapsto (4,4)$$
$$(7,7) \mapsto (3,3)$$



$\sigma' = \mathcal{F}^{-1}(\mathcal{Q}^2\theta) = 213469875$

RtL minima $(2,1), (3,3), (4,4), (9,5)$

Here $\theta = 451326798$ and $\mathcal{Q}^2\theta = 213487956$. The reflection ρ (the action of $\mathcal{D}$ on points) carries the RtL minima of σ *exactly* onto those of σ'.

Why ρ works (Lemmas 5.18, 5.19)

Split fixed points. $\mathcal{Q}^2$ rotates the diagram of θ by 180°, which swaps the NW and SE quadrants of (i, i). So $i \in S(\theta) \iff n{+}1{-}i \in S(\mathcal{Q}^2\theta)$. On the diagonal, the 180° rotation and ρ agree.

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Descent pairs. If θ has the descent $a\,b$ at positions $j, j{+}1$, then $\mathcal{Q}^2\theta$ has the descent $(n{+}1{-}b)\,(n{+}1{-}a)$ at positions $n{-}j, n{-}j{+}1$. So

$$\rho : E(\theta) \xrightarrow{\sim} E(\mathcal{Q}^2\theta).$$

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The $E^{\min}$ condition survives. ρ is a reflection across the anti-diagonal, which maps the SE quadrant of a point onto the SE quadrant of its image. So "no descent pair to my southeast" is preserved:

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Combining with Lemma 5.17: ρ restricts to a bijection between the RtL minima of $\mathcal{F}^{-1}\theta$ and those of $\mathcal{F}^{-1}\mathcal{Q}^2\theta$. □

Methodology

Where do the f's come from?

Linear algebra over a library of statistics. Fix a list $t_1, \dots, t_m$ of statistics (fix, exc, des, cyc, max, indicators like $[\pi(1) = n]$, ...) and look for

$$f = \sum_j a_j t_j \quad \text{with} \quad s(\pi) - c = \sum_j a_j \big(t_j(\pi) - t_j(\tau\pi)\big) \quad \forall \pi \in S_n.$$

That is $n!$ linear equations in m unknowns: solve by least squares, round, then re-check with exact rationals for several n.

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What comes out.

- A candidate f that holds for $n \leq 7$ or 8. Then a human proof is needed (cautionary tale ahead).
- When the library is too small there is no exact solution, and the library has to grow. For example, $[\mathrm{cyc}(\pi) = 1]$ is needed for two rows of the second table.
- The search was partly automated with AI assistance. Everything in the proved table has a human proof.

Telescoping functions proved in the paper

#	Foatic map	Statistic s	Telescoping f	c	Source
1–4	$\mathcal{C}$-$\mathcal{I}$, $\mathcal{C}$-$\mathcal{Q}^2$, $\mathcal{R}$-$\mathcal{Q}^2$, $\mathcal{R}$-$\mathcal{I}$	fix	$-\operatorname{exc}$	1	Thm 3.1
5	$\mathcal{Q}^2$-$\mathcal{C}$	$\overleftarrow{\max} - \overleftarrow{\min}$	$\overleftarrow{\max}$	0	Thm 3.2
6	$\mathcal{Q}^2$-$\mathcal{R}$	$D_1 - D_{n-1}$	$D_1 - D_1 \circ \mathcal{I}$	0	Thm 3.3
7–8	$\mathcal{R}$-$\mathcal{D}$, $\mathcal{D}$-$\mathcal{R}$	$\operatorname{fix}_1 - \operatorname{fix}_n$	$-[\pi(1){=}n] - \operatorname{fix}_n$	0	Thm 3.4
9–10	$\mathcal{C}$-$\mathcal{D}$, $\mathcal{D}$-$\mathcal{C}$	$\operatorname{fix}_1 - \operatorname{fix}_n$	$\operatorname{fix}_1 + [\pi(1){=}n]$	0	Thm 3.4
11–12	$\mathcal{C}$-$\mathcal{D}$, $\mathcal{R}$-$\mathcal{D}$	wexc	$\frac{1}{2}\operatorname{wexc}$	$\frac{n+1}{2}$	Sheridan-Rossi

Rows 11–12: Sheridan-Rossi proved $\operatorname{wexc}(T\pi) = n + 1 - \operatorname{wexc}(\pi)$, which is a telescope in disguise (Cor. 6.1).

Rows grouped together share an f, so each group is a family of compatible actions.

More telescoping functions (tentatively proved since the paper draft)

Map	Statistic	f	c
$\mathcal{C}$-$\mathcal{C}$	$\mathrm{fix}_1 - \mathrm{fix}_n$	fix_1	0
$\mathcal{R}$-$\mathcal{R}$	$\mathrm{fix}_1 - \mathrm{fix}_n$	$-\mathrm{fix}_n$	0
$\mathcal{Q}^2$-$\mathcal{C}$	$\mathrm{fix}_1 - \mathrm{fix}_n$	$-\mathrm{fix}_n - [\pi(n){=}1]$	0
$\mathcal{Q}^2$-$\mathcal{R}$	$\mathrm{fix}_1 - \mathrm{fix}_n$	$\mathrm{fix}_1 + [\pi(n){=}1]$	0
$\mathcal{I}$-$\mathcal{Q}^2$, $\mathcal{I}$-$\mathcal{D}$, $\mathcal{D}$-$\mathcal{I}$	$\mathrm{fix}_1 - \mathrm{fix}_n$	fix_1 (or $-\mathrm{fix}_n$)	0
$\mathcal{C}$-$\mathcal{Q}^2$	$\mathrm{fix}_1 - \mathrm{fix}_n$	$-\mathrm{fix}_n - [\mathrm{cyc}(\pi){=}1]$	0
$\mathcal{R}$-$\mathcal{Q}^2$	$\mathrm{fix}_1 - \mathrm{fix}_n$	$\mathrm{fix}_1 + [\mathrm{cyc}(\pi){=}1]$	0
$\mathcal{Q}^2$-$\mathcal{I}$, $\mathcal{Q}^2$-$\mathcal{Q}^2$	$\mathrm{wexc} + \mathrm{exc}$	exc	n
$\mathcal{Q}^2$-$\mathcal{I}$, $\mathcal{Q}^2$-$\mathcal{Q}^2$	$\underrightarrow{\max} - \underleftarrow{\min}$	$\underrightarrow{\max}$	0

Patterns worth noticing.

- Each $\mathrm{fix}_1 - \mathrm{fix}_n$ row is a few-line chain: track one entry through $\mathcal{F}$, $\mathcal{A}$, $\mathcal{F}^{-1}$, $\mathcal{B}$.
- New compatible families: $\mathcal{C}$-$\mathcal{C}$, $\mathcal{I}$-$\mathcal{Q}^2$, $\mathcal{I}$-$\mathcal{D}$, $\mathcal{D}$-$\mathcal{I}$ via fix_1; and $\mathcal{R}$-$\mathcal{R}$, $\mathcal{I}$-$\mathcal{Q}^2$, $\mathcal{I}$-$\mathcal{D}$, $\mathcal{D}$-$\mathcal{I}$ via $-\mathrm{fix}_n$.
- The $\mathcal{Q}^2$-first maps $\mathcal{Q}^2$-$\mathcal{B} = \mathcal{B} \circ U$ keep reappearing. Is there a structural theory of $U = \mathcal{F}^{-1}\mathcal{Q}^2\mathcal{F}$?
- The last two rows reduce to Prop. 5.8 again. Contrast $\underrightarrow{\max} - \underleftarrow{\min}$ under $\mathcal{I}$-$\mathcal{I}$, which fails at $n = 9$.

Non-uniqueness is informative

For $\mathcal{I}$-$\mathcal{Q}^2$, $\mathcal{I}$-$\mathcal{D}$, $\mathcal{D}$-$\mathcal{I}$, *two* telescoping functions work for $\mathrm{fix}_1 - \mathrm{fix}_n$:

$$f_1 = \mathrm{fix}_1 \qquad \text{and} \qquad f_2 = -\mathrm{fix}_n\,.$$

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By Remark 4.2, two telescoping functions differ by a function constant on orbits:

$$g := f_1 - f_2 = \mathsf{fix}_1 + \mathsf{fix}_n \quad \text{satisfies} \quad g(\tau\pi) = g(\pi).$$

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$$g := f_1 - f_2 = \mathrm{fix}_1 + \mathrm{fix}_n \quad \text{satisfies} \quad g(\tau\pi) = g(\pi).$$

So we get a new invariant for free (given the tentative proofs): for these three actions, the number of fixed points among the two end positions never changes along an orbit.

In general, any discrepancy between two fitted f's is an orbit invariant, which can be useful for understanding the orbit structure.

A cautionary tale: small n can lie

Both statistics below are homomesic under $\mathcal{I}$-$\mathcal{I}$ for every $n \leq 8$, but fail at $n = 9$:

k	$\tau^k(\pi)$	wexc + exc
0	258673491	12
1	923678451	10
2	915672438	8
	average	$10 \neq 9$

k	$\tau^k(\pi)$	$\underrightarrow{\max} - \underleftarrow{\min}$
0	246738591	5
1	923784651	0
2	914723568	-5
3	267843591	4
4	923754861	0
5	916732548	-3
	average	$1/6 \neq 0$

- Compare Elder et al.: 387 FindStat statistics $\times$ 19 bijections, tested for $2 \leq n \leq 6$. "We did not find any false positives when using the data from $n = 6$"; all 117 candidates were then proved.
- Foatic maps were *not* in that sweep: they are compositions of FindStat maps, not single entries.

How sparse are the failures?

All orbits of $\mathcal{I}$-$\mathcal{I}$ on S_n (exhaustive search):

n	orbits	largest orbit	bad orbits for wexc + exc	bad orbits for $\overrightarrow{\max} - \overleftarrow{\min}$
4	10	5	0	0
5	26	27	0	0
6	78	67	0	0
7	242	169	0	0
8	828	691	0	0
9	2908	2354	10 (sizes $3, 6, 9, 20$)	6 (sizes $6, 9, 20$)

- At $n = 9$ the failures cover 88 (resp. 70) of 362,880 permutations, so the orbit average is correct on more than 99.6% of orbits.
- Small orbits are *not* rare: 1172 of the 2908 orbits have size ≤ 20. So the failures are not hiding in exotic orbits. They are just very few.

Why might Foatic maps fool small-n testing?

Some possible heuristics:

- **Structured vs. unstructured orbits.** Most of Elder et al.'s homomesies live on maps with explicit orbit structure: $\mathcal{C}$ and $\mathcal{R}$ are involutions, Kreweras has order $2n$, and Lehmer code rotation has all orbits of size $\mathrm{lcm}(1, \ldots, n)$. Homomesy there tends to come from a uniform mechanism that is already visible at small n. Foatic orbits look irregular, so nothing forces a coincidence to break early.

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- **Telescoping as a diagnostic.** Search the span of 20 standard statistics (fix, exc, des, inv, maj, cyc, LtR/RtL extrema, end indicators, ...) for a telescoping f, for $n = 5, \ldots, 8$:

fix under $\mathcal{C}$-$\mathcal{I}$ (true)	exact fit, $f = -\mathrm{exc}$, every n
$\mathrm{wexc} + \mathrm{exc}$ under $\mathcal{I}$-$\mathcal{I}$ (false at 9)	no fit at any n, even where it holds
$\underrightarrow{\max} - \underleftarrow{\min}$ under $\mathcal{I}$-$\mathcal{I}$ (false at 9)	no fit at any n

  Rule of thumb: a homomesy with no simple telescope is a red flag.

The rule of thumb can misfire

An engineered true homomesy. Take an "unnatural" function and *define* the statistic from it:

$$f(\pi) := \pi(1) + \pi(2), \qquad s := f - f \circ \tau.$$

By Lemma 4.1, s is 0-mesic under τ for every n. But the library search finds no telescope:

best library fit, max error	$n = 5$	$n = 6$	$n = 7$	$n = 8$
$\tau = \mathcal{I}$-$\mathcal{I}$	0.71	2.70	4.69	6.81
$\tau = \mathcal{C}$-$\mathcal{I}$	0.86	2.38	4.55	6.83

So "no telescope in the library" cannot by itself mean "false".

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best library fit, max error	$n = 5$	$n = 6$	$n = 7$	$n = 8$
$\tau = \mathcal{I}$-$\mathcal{I}$	0.71	2.70	4.69	6.81
$\tau = \mathcal{C}$-$\mathcal{I}$	0.86	2.38	4.55	6.83

So "no telescope in the library" cannot by itself mean "false".

But here s is unnatural too. It is not in the span of the library either. A sharper rule: *a natural s that is homomesic for small n but has no natural telescope is suspicious.*

Question. Is there a natural statistic that is homomesic under a natural map but whose only telescopes are unnatural?

The other direction: natural f and τ, semi-natural s

Start from the telescope: take $\tau = \mathcal{Q}^2\text{-}\mathcal{C}$, $f = -\mathrm{exc}$, $c = 1$, and *define* $s(\pi) := 1 - \mathrm{exc}(\pi) + \mathrm{exc}(\tau\pi)$, which is 1-mesic under $\mathcal{Q}^2\text{-}\mathcal{C}$ for every n. s is not in the library's span (best-fit error 1.3 at $n = 5$, growing to 2.9 at $n = 8$). But it can be read off the CCD without computing τ:

1. Write the CCD of π and drop the parentheses to get the word $w = \mathcal{F}(\pi)$.
2. Mark each letter smaller than everything to its right, and cut w just after each mark.
3. $s(\pi) = \mathrm{fix}(\pi) - \mathrm{asc}(w) + \#\{\text{adjacent } xy \text{ inside a piece} : x + y > n + 1\} + \#\{\text{pieces with first} + \text{last} > n + 1\}$.

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Example. $\pi = (5,4)(6,3)(7,1,2)$: $w = 546371\,|\,2$, so $s = 0 - 3 + 4 + 0 = 1$.

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Why. $\mathcal{F}^{-1} \circ \mathcal{Q}^2 = \mathcal{D} \circ G^{-1}$, where G drops parentheses in the *min-last* cycle convention (each cycle ends with its minimum, cycles sorted by increasing minimum). So $\tau\pi = \mathcal{R}\mathcal{I}\,G^{-1}(\mathcal{F}\pi)$, and the last two terms count the arcs of $G^{-1}(\mathcal{F}\pi)$ above the anti-diagonal.

So the rule of thumb also cuts the other way: natural telescopes can certify homomesic statistics that no reasonable library would contain.

Open questions

1. **Understand $U = \mathcal{F}^{-1} \circ \mathcal{Q}^2 \circ \mathcal{F}$.** Prop. 5.8 says U preserves $\underleftarrow{\min}$, and this drives Theorem 3.2 and the last two rows of the second table. Which other statistics does U preserve?
2. **Natural statistics, unnatural telescopes?** Is there a natural statistic, homomesic under a natural map, whose only telescopes are unnatural?
3. **Fixed points of $\mathcal{I}$-$\mathcal{I}$.** $\tau\pi = \pi \iff \mathcal{F}(\pi^{-1}) = \mathcal{F}(\pi)^{-1}$. For $n = 1, \ldots, 9$ there are

   $$1, 2, 3, 5, 8, 13, 21, 34, 55 \qquad \text{(Fibonacci)}$$

   of them, and (checked for $n \leq 9$) they are exactly the products of disjoint *adjacent* transpositions, e.g. $13254 = (1)(3,2)(5,4)$. These clearly work, since $\mathcal{F}\pi = \pi = \pi^{-1}$ (they form $S_n(231, 312, 321)$, counted by F_{n+1}: Simion–Schmidt 1985, Prop. 15 with Lemma 6(a)). Are there no others?

Next direction: telescoping families

Fix a statistic s, a constant c, and a telescoping function f. Which actions τ satisfy

$$s = f - f \circ \tau + c \quad \Longleftrightarrow \quad f \circ \tau = f - s + c \ ?$$

- If τ_0 is one solution, the set of all solutions is $\{\sigma \circ \tau_0 : f \circ \sigma = f\}$. That is huge, so the real question is which solutions are natural.
- Every solution is s-compatible with every other, so the whole family shares the orbit-average and closed-walk properties.

Thank you!

Questions welcome.

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Thanks to Darij Grinberg for suggesting the telescoping approach.
AI tools were used to find candidate telescoping functions from data and to computer-check proof steps (see the paper).

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